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Modeling Interface Debonding in Coated Fuel Particles with BISON

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Abstract — *This work introduces a meshed debonding model intended for use with tristructural isotropic (TRISO) fuel particles. The ability to better examine the potential effects of interface debonding between the coating layers will enable a host of investigations of the stress state when complex loading and interactions between coated fuel particles are taking place. In this work, the debonding model is discussed in detail. Then, simulations are presented to show the capabilities and limitations of the model. First, a traditional high-temperature gas reactor TRISO particle is considered. Then, a TRISO particle in a graphitic matrix is analyzed, followed by a TRISO particle in a silicon carbide matrix.*

Keywords — *TRISO, debonding, silicon carbide pyrolytic carbon.*

Note — *Some figures may be in color only in the electronic version.*

I. INTRODUCTION

Coated fuel particles have been under development and use for many decades with the tristructural isotropic (TRISO) fuel form providing exceptional ability to retain radionuclides and reach high burnups.[1,2] The coated fuel particle generally consists of a uranium-bearing kernel at the center that is subsequently coated with a porous buffer and a pyrolytic carbon layer, respectively, to form a bistructural isotropic (BISO) particle. If an additional carbide ceramic coating layer followed by an outer pyrolytic carbon (OPyC) layer is included, then the TRISO particle is formed. The kernel materials may be oxides, carbides, or nitrides of uranium or their combinations,[2–5] and the ceramic carbide coating layer is often silicon carbide[6] (SiC), while other carbides and nitrides have also been produced previously.[7–9]

These distinct constituents of the coated fuel particle possess different temperature-dependent thermophysical properties and are affected by a diverse set of mechanisms under displacement-induced irradiation damage. The result is that upon heating, cooling, and burnup accumulation, these distinct materials experience differential strains, and based on their mechanical properties, they experience different states of stress. These stresses are imposed on the various constituents as well as the interfaces between them. Interfacial stresses may result in debonding that will mechanically decouple these constituents and greatly affect the stress distribution within the coated fuel particle.

Debonding of the layer interfaces and the associated effects on the mechanical and failure behavior of TRISO particles have been observed experimentally.[10–12] For instance, the AGR-1 experimental campaign[11] studied the performance of UCO (uranium carbide, uranium oxide) TRISO particles during irradiation in Idaho National Laboratory's Advanced Test Reactor and postirradiation safety testing at high temperature. Microscopy analysis of the irradiated particles revealed the occurrence of debonding, having important relationships with layer fractures. The typical behavior observed was debonding between the buffer and the inner pyrolytic carbon (IPyC) layer, resulting from irradiation-induced buffer densification. Buffers that debonded from the IPyC interior could not locally concentrate stresses in the IPyC, resulting in no cracking

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of the IPyC and SiC layers found in particles with complete buffer-IPyC separation.[11]

Conversely, IPyC-SiC debonding occurred in particles where the buffer/IPyC interface was entirely bonded and was determined to adversely affect the propensity to cracking of the SiC layer. In particular, cracks in the SiC layer formed in the regions where the SiC was exposed and palladium was allowed to concentrate. In some cases, radial cracks through the entire SiC layer thickness were observed in particles with IPyC-SiC debonding.[11] Furthermore, during postirradiation safety testing, significant release of Cs was observed from particles that experienced SiC failure. OPyC debonding may also occur during irradiation if shrinkage in the matrix takes places.[11]

These experimental results highlight the importance of debonding in the mechanical behavior of TRISO particles, including the failure propensity of the SiC layer and its implications in terms of release of radioactive fission products. It follows that including reliable models for layer interface debonding in fuel performance codes is necessary to perform accurate simulations of TRISO particle fuel behavior.

Development and assessment of a computational methodology to appropriately implement debonding and stress redistribution across the various interfaces in the coated fuel particles are the subject of this paper.

II. METHODOLOGY

The simulations presented in this work were performed using the BISON fuel analysis tool.[13,14] Since application of this tool to TRISO fuel particles has been discussed in detail previously,[13,15] the focus of this work is on details of approaching and resolving interface debonding for the TRISO particle coating layers.

The approach pursued here is application of a meshed debondable interface. The high-level overview of this model is that the PyC layers (inner and outer, IPyC and OPyC) are subdivided into the main layer, which will always have intact PyC properties, and a thin debondable region between the main PyC layer and the adjacent constituents (e.g., SiC coating layer). Once the criterion for debonding, discussed in Secs. II.A, II.B, and II.C, is met for an individual element in this debondable region, the elastic modulus of that element is greatly reduced to allow for the element to deform with essentially no induced stress in the PyC or SiC layers. This allows for a pseudoseparation of the coating layers. Figure 1 shows a general IPyC (brown), SiC (green), and OPyC (blue) geometry with debondable regions (red) between the PyC layers and SiC.

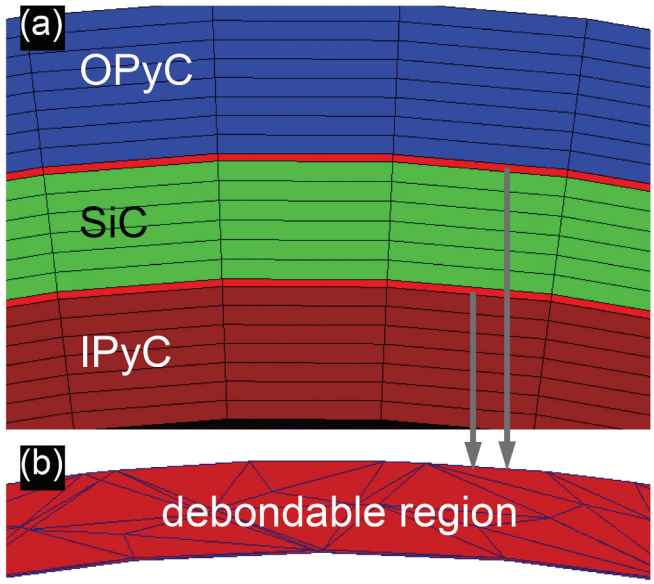


Fig. 1. Geometry setup showing the SiC layer between IPyC and OPyC with (a) single element debondable region in red to be used with stress-based debonding criterion and (b) zoomed-in view of meshed debondable region that requires a strain-based debonding criterion.

In this work, stress-based and strain-based debonding criteria are both considered. The stress-based criterion uses a stress threshold to initiate debonding and is well suited for general use. However, it requires that the debondable region between the layers must be only one element thick, Fig. 1a. Extensive convergence issues have been encountered if the debondable region is multiple elements thick when using the stress-based criterion. However, the strain-based criterion uses a strain threshold that functions well with multi-element-thick debondable regions. Consideration of a meshed debondable region may be desirable when assessing complex stress states in three dimensions. However, in certain situations the available meshing techniques can only produce a complex multielement debondable region similar to that shown in Fig. 1b. In those specific cases the strain-based debonding criterion is required. However, this alternate method assumes that the layer is fully constrained by a hard material such as SiC.

II.A. Stress-Based Model Description

Weibull statistics, shown in Eq. (1), are employed to estimate the probability of an element debonding for the simulated geometries based on prior experimentally determined parameters.

The experimental parameters determined from material testing include the characteristic volume V_c of the samples that were tested in the experimental campaign and the determined characteristic normal stress and shear stress σ_c and τ_c, respectively, and Weibull modulus m. The variables calculated during the simulation are the mesh element volume V_e as well as the predicted radial stress and maximum shear stress within that element σ_r and τ_{max}, respectively. The failure probability from normal stress and shear stress are calculated separately and then added together. Normalization of the mesh element volume to the characteristic volume is critical in the correct application of Weibull statistics because crack nucleation occurs at flaws in the ceramic materials. As the volume increases, the mean fracture stress in the region decreases due to the increasing probability of a critical flaw being present, and vice versa. Equation (1) gives the relationship for the total failure probability per element:

$$F_P = 2 - EXP\left[\frac{-V_e}{V_c}\left(\frac{\sigma_r}{\sigma_c^{normal}}\right)^m\right] - EXP\left[\frac{-V_e}{V_c}\left(\frac{\tau_{max}}{\tau_c^{shear}}\right)^m\right] . \quad (1)$$

Table I lists the material properties used in the model to calculate the probability of interface debonding for PyC. The characteristic normal stress value of 100 MPa is the upper bond strength determined from comparisons with TRISO irradiations during the Non-Power Reactor (NPR) program.[16] For the buffer material, a value of 5 MPa was estimated as the characteristic strength of the buffer-IPyC bond to prevent the buffer shrinkage from inducing significant stresses in the outer coating layers. It is recognized that although these bond strengths may vary from what is assumed here, the overall methodology is unaffected and may be applied to other values.

The debonding criterion for each element is established at the beginning of the simulation. Each element is assigned a random value between 0 and 1 with a uniform distribution. If the calculated probability of debonding exceeds this debonding criterion, then the element is considered eligible for debonding. At this point, the elastic modulus of the element eligible for debonding is reduced to a small fraction of its original value.

TABLE I

A List of the Parameters Used for Calculating the Debonding of the PyC Layers

Parameter	Value
PyC characteristic volume (m^3)	2.25×10^{-12}
PyC characteristic shear stress (MPa)	240
PyC characteristic normal stress (MPa)	100
PyC Weibull modulus	9.5

*References 17, 18, and 19.

II.B. Strain-Based-Criterion Description

As previously discussed, the strain-based criterion is used only when a complex mesh in the debondable region is unavoidable. This criterion is similar to the previous stress-based criterion except that the failure probability is estimated using the volumetric strains from thermal expansion and irradiation-induced dimensional change. Equation (2) shows how the failure probability is calculated in the strain-based debonding model:

$$F_p = 1 - EXP\left[\frac{-V_e}{V_c} \times \left(\frac{\sim\sigma_r}{\sigma_c^{normal}}\right)^m\right] \quad (2a)$$

and

$$\sim\sigma_r = -\frac{E}{(1+\upsilon)(1-2\upsilon)} [(1-\nu)(AS_r + B\varepsilon_r) + \nu(CS_T + D\varepsilon_T)] , \quad (2b)$$

where

F_p = failure probability of the element
V_e = volume of the element
V_c = characteristic volume from material testing
m = Weibull modulus
$\sim\sigma_r$ = estimated radial stress (Pa)
E = elastic modulus (Pa)
υ = elastic Poisson's ratio
S_r = radial dimensional change
S_T = tangential dimensional change
ε_r = radial creep strain
ε_T = tangential creep strain
A, B, C, D = constants.

The probability and the occurrence of debonding in any individual element and the reduction in that element's modulus are treated similarly to what is described in Sec. II.A.

II.C. Gap Thermal Conductivity

The effective thermal conductivity of a debonded element is derived from one-dimensional heat transfer and makes the conservative assumption that the entirety of the heat must pass through the gap. The debonded elements are a combination of gas and PyC. The composition of the gas mixture in the gaps between layers is calculated based on the fission product diffusion and release models available in BISON (e.g., Ref. 13). The thermal conductivity of the gases was determined as a function of temperature using MATPRO equations[20] contained in BISON material files. The volume fraction of the gap was derived from the elastic strains. The model does not include radiative heat transfer across the gap.

Equation 3a presents the approximation for the average thermal conductivity of a debonded element, and Eq. 3b presents the thermal conductivity of the gas mixture:

$$\bar{k} = \frac{1}{\frac{1-gapFraction}{k_{\mathrm{PyC}}} + \frac{gapFraction}{k_{gas}}} \tag{3a}$$

and

$$k_{gas} = \sum k_i X_i \quad , \tag{3b}$$

where

$\bar{k}$ = average element conductivity
k_{PyC} = thermal conductivity of the intact PyC
k_{gas} = estimated conductivity of the gas mixture contained in the gap
$gapFraction$ = estimated fraction of the element volume that contains the gap
k_i, X_i = conductivity and molar fraction of each gas, respectively.

Figure 2 plots the resulting element average thermal conductivity versus the gap volume fraction for a debonded hydrogen-PyC element at 873 K. The resulting thermal conductivity trend follows a reciprocal behavior with increasing gap volume fraction. This predicts that the thermal conductivity of the element rapidly decreases during the initial crack opening and subsequently becomes less sensitive to additional crack volume. With that said, the detailed assessment of gas composition in the gap and thermal transport mechanisms is outside the scope of the present work and does not affect the proposed methodology.

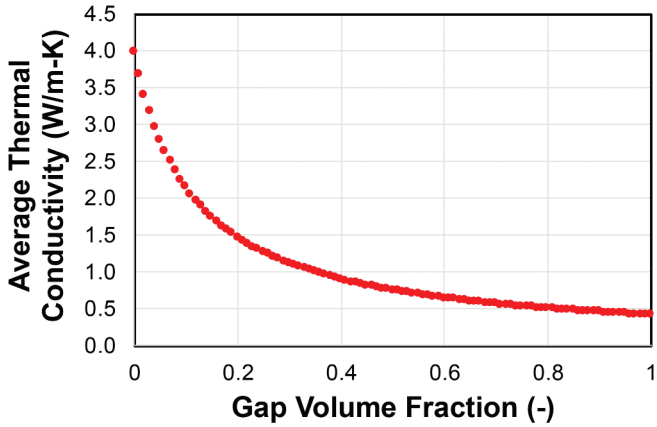


Fig. 2. Plot of the element average thermal conductivity of PyC as a function of gap volume fraction. For this example calculation, the gas in the gap is assumed to be H_2 at 1 atm.

III. RESULTS

III.A. Basic Debonding and Gap Width Evolution

The following simulation was designed to assess the efficacy of the stress-based debonding model and the effect of reducing the elastic modulus in the debondable region. The simulation geometry is shown in Fig. 3a. The central green sphere is assumed to consist of carbon buffer and is set to shrink by about 30%. The white

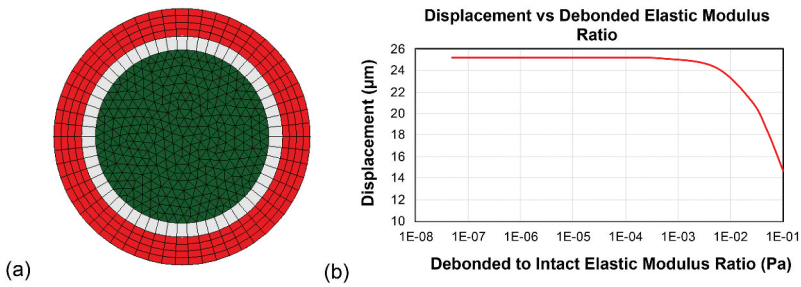


Fig. 3. (a) Test case configuration. (b) Numerical gap width versus modulus ratio.

shell is the debondable region, and the red shell has PyC properties. For this debonding simulation, mesh elements 10 μm wide in the radial direction were assumed and found to produce satisfactory results.

The green sphere shrinks by 27.1% volumetrically (~10% in radius) as the simulation progresses, which opens the gap between the center sphere and the outer red shell. The results plotted in Fig. 3b show that the predicted displacement saturates for debonded-to-intact elastic modulus ratios less than about 5×10^{-4} and remains reasonably unchanged up to the ratio of 5×10^{-3} (0.5%), which is the reduced value of the elastic modulus once debonding is implemented. The difference between the predicted displacements in BISON and the analytical calculations were 1.1% for a debonded modulus ratio of 5×10^{-4} and 2.4% for a debonded modulus ratio of 5×10^{-3}. The analytical displacement is calculated from the volumetric change of the green sphere and the red shell. Since the outer surface of the red shell is fully constrained, the change in radius between the inner surface of the red shell and the green sphere can be calculated via simple arithmetic assuming a fully decoupled interface. These changes in radius produce the correct gap width that can be used to verify the BISON results.

III.B. Simulation of Free TRISO Particle at High Temperature

The conditions for these simulations are from the International Atomic Energy Agency Case 11 benchmark.[21] The simulated time is 351 days, at a volume-averaged temperature of 1608 K, maximum fast neutron fluence of 7.2×10^{25} n/m^2, and a burnup of 14% fissions per initial metal atom (FIMA). In these simulations, the effect of debonding between the SiC and IPyC layer was considered. The study shows how the debonding model performs as a gap is predicted to form and subsequently close due to the increasing gas pressure at the inner surface of the IPyC layer. The predicted gas pressure was about 67 MPa at the end of the simulation. Figure 4a shows the simulated geometry. The two red layers are the debondable regions, which are the IPyC/SiC and buffer/IPyC. The blue is OPyC, green is SiC, gray is IPyC, purple is buffer, and yellow is the fuel kernel. Figure 4b plots the tangential stress on the inner edge of the SiC layer. In both cases the strength of the buffer/IPyC interface was set to 5 MPa. This allows for the buffer/IPyC interface to debond without inducing appreciable stress in the outer coating layers. The debonded modulus of the buffer/IPyC debondable region was set to 1 kPa while the debonded modulus of the IPyC/SiC debondable region was set to 10 MPa.

Initially, the SiC shell is placed in compression by the shrinkage of the OPyC and IPyC layers. However, as the shrinkage rate slows, the PyC creep relaxes the stress. Then, as the gas pressure increases, the stresses become increasingly tensile. When the bond strength between the IPyC and SiC is set to 10 MPa and debonding is predicted early in the simulation, the resulting initial compressive tangential stresses are smaller due to there being less material pulling the SiC layer inward. However, as the internal gas pressure increases, the IPyC creeps back into contact with the SiC layer. The scatter in the stress trend for fluences between about 4×10^{25} to 5×10^{25} n/m^2 is due to this recontact process. After firm contact is reestablished, the stresses return to a consistent trend.

III.C. Simulation of TRISO Inside a Graphitic Matrix

A set of simulations was performed to show the impact of debonding between the SiC and OPyC coating layers in a graphitic matrix. The geometry consists of a three-dimensional TRISO particle in a graphitic matrix. Two interfaces in the TRISO particles were permitted to debond in these simulations, which are the buffer/IPyC and the SiC/OPyC interfaces. The other interfaces, including the OPyC/matrix interface, were constrained to remain in contact. Figure 5 shows the geometry, where the two bright red

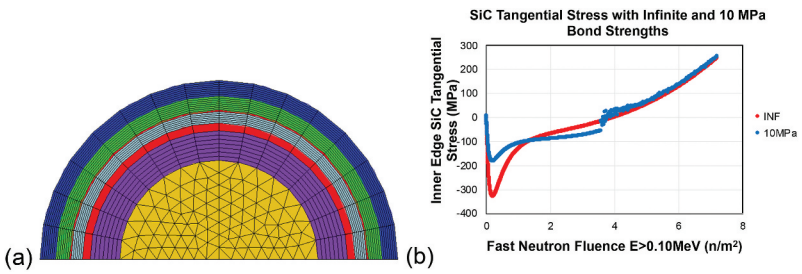


Fig. 4. (a) Simulated geometry for a free TRISO particle. (b) Plot of the tangential stress on the inner edge of the SiC coating layer with SiC/IPyC bond strength set to infinity or 10 MPa.

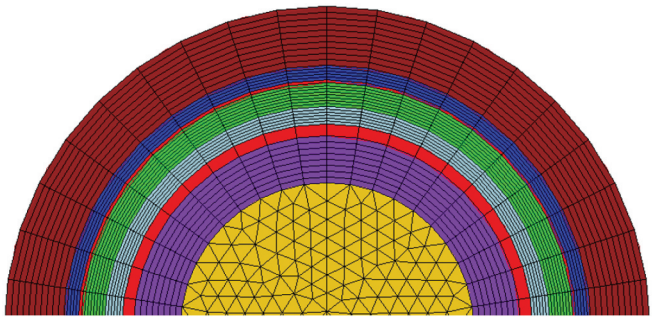


Fig. 5. Simulation geometry for a TRISO particle in a graphitic matrix.

areas are the debondable regions. The dark red/brown area is the matrix, the blue is the OPyC, the green is the chemical vapor deposited (CVD)-SiC, the gray is the IPyC, the purple is the buffer, and the yellow is the fuel kernel.

The simulation conditions consisted of a steady-state irradiation at 873 K, which is almost half the value of the previous simulation. This low temperature was selected so that internal gas pressure would not have a significant effect in these simulations. The irradiation-induced dimensional change (IIDC) of the PyC consists of tangential shrinkage and radial expansion with a net volumetric reduction. While the SiC layer is expected to swell by about ~1% at these temperatures, because of these simulated conditions, the OPyC will shrink onto the SiC creating a compressive normal stress and small amounts of shear stress. This simulation shows that the model can converge in the case where there is compressive normal stress, but the shear stress is allowed to debond the interface. This ability becomes essential for the situation where two separated layers are in very close proximity to each other.

In this simulation, the characteristic shear stress was set to an unrealistically small 0.5 MPa. The simulation does not predict the formation of a gap between the OPyC and SiC layers but rather some amount of slippage between the layers. This slippage results in the scatter shown in the blue data in Fig. 6, which compares the results of infinite and 0.5-MPa bond strengths on the SiC/OPyC interface.

III.D. Simulation of TRISO Inside a SiC Matrix

This section repeats the previous graphitic matrix simulations using the same geometry and conditions, but the graphitic matrix has been replaced with a SiC matrix. The SiC matrix has a similar behavior to the SiC coating layer, which is about 1% swelling and negligible creep. The stress in the case with infinite bond strength is dominated by the predicted volumetric behavior of the IIDC in the OPyC. This is possible because the simulation has unrealistically constrained the OPyC to not crack and remain in contact with both the SiC coating layer and the SiC matrix. Also, irradiation creep in the simulation was modeled as occurring at constant volume. Using this set of constraints will cause the prediction of very large stresses.

Subsequent simulations reduce the normal bond strength from 200 to 10 MPa. The results of these simulations show that allowing the SiC and OPyC layers to effectively separate greatly reduces the predicted stresses. Figure 7 shows plots of the maximum stresses in the SiC coating layer and the matrix. While the 200-MPa bond strength is unrealistically high, it does show that the maximum radial stress is predicted to increase somewhat during the debonding process. In the brief time when part of the interface is predicted to have debonded while the remainder is still in contact, the stress is concentrated, which leads to a bubble in the maximum stress trend. However, once the interface has completely debonded, the maximum stress is greatly decreased. The stress trends after the SiC/OPyC interface has completely debonded are essentially identical for all bond strengths as expected.

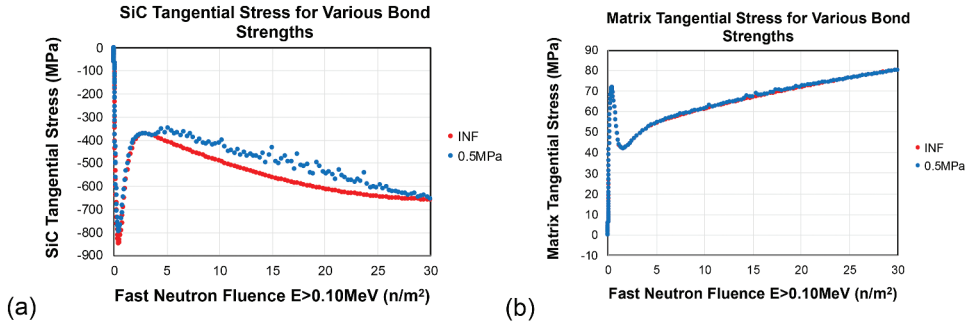


Fig. 6. Plots of the tangential stress in (a) the SiC coating layer and (b) the graphitic matrix.

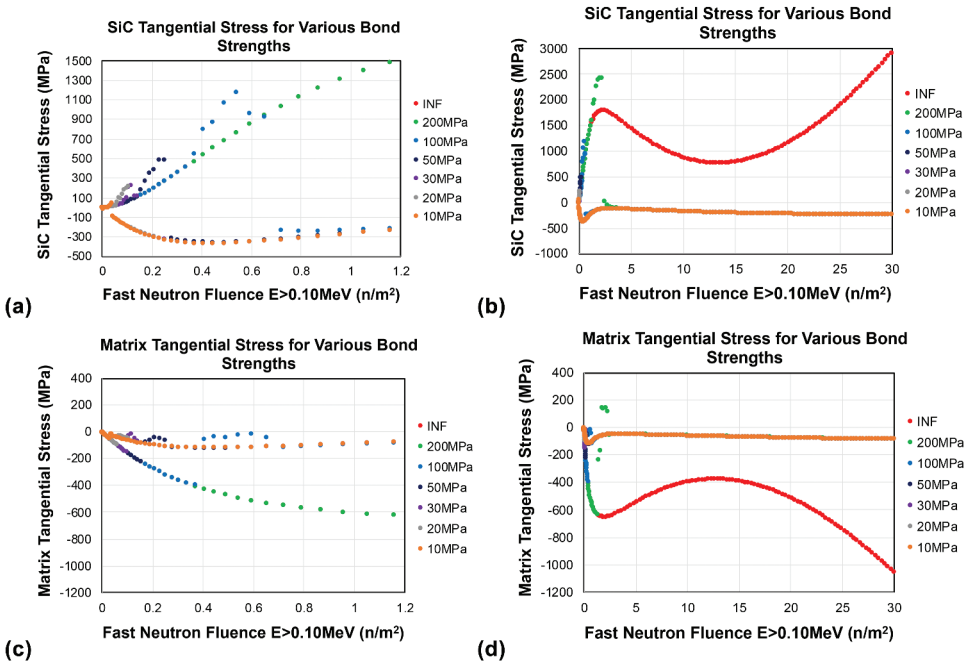


Fig. 7. (a) and (b) Plots of the maximum tangential stress in the SiC coating layer. (c) and (d) Plots of the maximum tangential stress in the matrix.

III.E. Simulation of TRISO Inside a SiC Matrix with Strain-Based Debonding

This set of simulations repeats those from Sec. III.D; however, the strain-based debonding criterion was selected instead of the stress-based criterion. Figure 8 plots the maximum stresses in the SiC coating layer. The strain-based debonding criterion predicts a smaller stress increase under partial debonding, and the debonding process occurs at lower stresses compared to the stress-based debonding criterion. However, the general effects are still predicted, and the trends are similar to those predicted by the stress-based debonding criterion.

IV. DISCUSSION

Section III.A verifies the predicted gap widths. It was found that the ratio of the debonded-to-intact elastic modulus ratio was the important factor, instead of the absolute value of the debonded elastic modulus. Various elastic moduli were selected for the outer red shell and green sphere. The model was found to perform well in all of these combinations as long as the debonded elastic modulus was at least a factor of 5×10^{-3} or smaller compared to the softer of the two materials.

The free particle simulations in Sec. III.B show the ability of the debonding model to predict the formation of a gap and subsequent closure of that gap. This capability could also be used for buffer/IPyC or kernel/buffer recontact. The scatter in the maximum stress on the inner edge of the SiC is due to the model using an explicit criterion for the recontact. Implicit forms can be used in small simulations but often have significant convergence problems in multiparticle simulations.

The TRISO inside a graphitic matrix presented in Sec. III.C shows the model predictions for the case where the OPyC is shrinking onto the SiC layer. However, there is a considerable amount of shear stress at the interface. Also, the local bond strength varies along the interface owing to the Weibull treatment. The model allows the interface to experience localized slippage, but the layers do not separate, and the simulation continues to converge. The scatter in the maximum stress trend is the result of one element having a weaker bond strength while a stronger nearby element experiences a stress concentration. This general capability has the potential to assist in investigating the formation of through-wall cracks.

The effect of a SiC matrix is presented in Sec. III.D. These simulations show that if the OPyC layer is constrained to remain in contact with both the CVD-SiC layer and the SiC matrix, then large stresses are predicted. However, the bond strength of SiC/PyC interfaces is less than their fracture strengths. Additionally, the model assumes that the Poisson ratio in creep for PyC is always

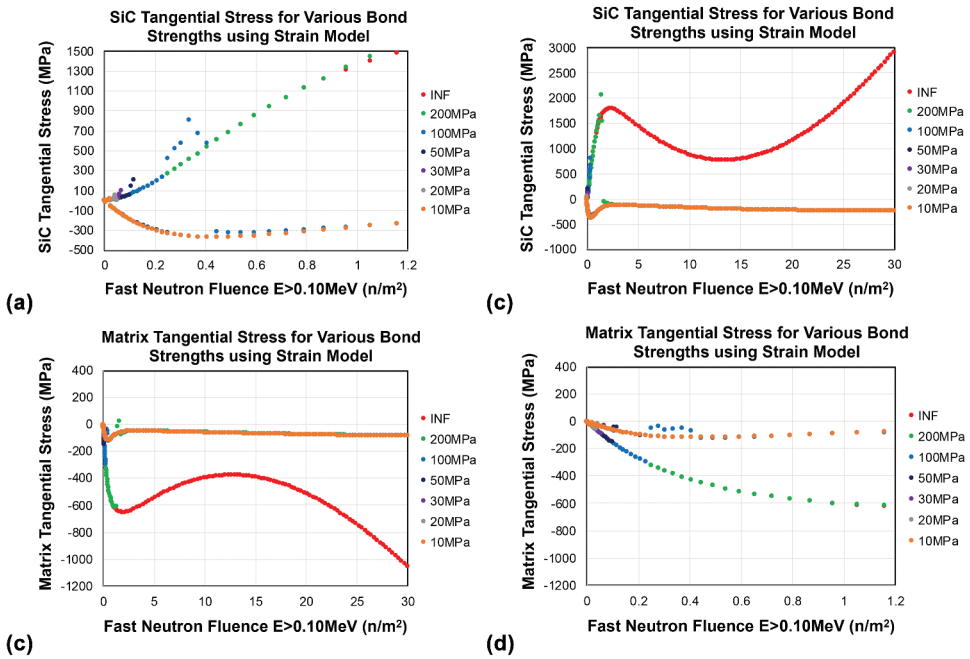


Fig. 8. (a) and (b) Plots of the maximum tangential stress in the SiC coating layer. (c) and (d) Plots of the maximum tangential stress in the matrix.

0.5, which conserves volume, and it also assumes that the IIDC behavior is unaffected by hydrostatic stress. This work allows for an investigation of the effect of allowing the SiC/OPyC interface to debond and indicates that a low bond strength between the SiC and OPyC layers is beneficial for TRISO particles in a SiC matrix. It also shows that if the OPyC is not unrealistically constrained, then TRISO particles in a SiC matrix should perform well.

In Sec. III.E the strain-based debonding model was put to test for a similar scenario assessed in Sec. III.D. Similar results were deduced form from both models.

V. SUMMARY

A mesh-based interface debonding model was created for TRISO simulations in BISON. It is important to set the debonded elastic modulus of the debondable region to a factor of at least 5×10^{-3} or less of its intact elastic modulus to achieve good results. The thermal conductivity of a debonded element is also reduced to estimate the effective thermal resistance of a gap between coating layers.

A simulation of a free particle at typical HTGR temperatures showed that the model can allow an interface to debond and subsequently reestablish contact. The simulation of a TRISO particle in a graphitic matrix shows that the model can continue to converge while allowing interface slippage in a compressive condition, which can result in element inversion.

The simulations of a TRISO particle in a SiC matrix show that in some cases allowing an interface to separate can make the constraints more representative and significantly impact the stress distributing in the TRISO coating layers. The simulations with the strain-based criterion show that it can deliver similar results if the fully constrained assumption of the OPyC is maintained.

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